

Detecting Phishing Websites Using Neural Networks

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COSC 50523: Deep Learning

Path A: Applied Deep Learning

1. Problem

Phishing websites imitate legitimate websites to trick users into revealing sensitive information. Because phishing websites can change rapidly, automated detection is an important cybersecurity problem^[2]. This project will investigate whether a neural network can accurately classify websites as phishing or legitimate using features extracted from their URLs and webpages.

2. Dataset

We will use the PhiUSIIL Phishing URL Dataset from the UCI Machine Learning Repository. The dataset contains labeled phishing and legitimate websites with extracted URL and webpage features^[1]. It is publicly available and can be accessed before implementation begins. The data will be divided into training, validation, and test sets. Numerical features will be scaled as needed, and preprocessing will be performed without using information from the test set.

Dataset: PhiUSIIL Phishing URL Dataset

Link: <https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+website>

3. Neural Approach and Baseline

We will use a Multilayer Perceptron (MLP) for binary classification. The MLP will take the extracted website features as input and predict whether a website is phishing or legitimate^[3]. Logistic Regression will serve as the baseline. Both models will use the same data splits and input features, allowing their performance to be compared directly. We will also conduct a small controlled experiment by varying an MLP design choice, such as the number of hidden layers or neurons.

4. Evaluation

Models will be evaluated on the test set using accuracy, precision, recall, and confusion matrices. Particular attention will be given to false negatives because incorrectly classifying a phishing website as legitimate can pose a security risk. Success will be evaluated by comparing the MLP with the Logistic Regression baseline and analyzing the results and representative classification errors.

5. Feasibility

The project is feasible within the semester because the dataset is publicly available and already contains extracted website features, eliminating the need to collect or interact with live websites. The proposed models can be trained using standard machine-learning and deep-learning libraries on available computing resources. The project will maintain reproducible code, documented preprocessing, model configurations, and experiment results.

References

- [1] A. Prasad and S. Chandra, “PhiUSIIL: A diverse security profile empowered phishing URL detection framework based on similarity index and incremental learning,” *Computers & Security*, vol. 136, Art. no. 103545, Jan. 2024, doi: 10.1016/j.cose.2023.04.003.
- [2] H. Ghalechyan, E. Israyelyan, A. Arakelyan, G. Hovhannisyan, and A. Davtyan, “Phishing URL detection with neural networks: an empirical study,” *Scientific Reports*, vol. 14, Art. no. 25134, Oct. 2024, doi: 10.1038/s41598-024-74725-6.
- [3] A. Rawla, S. Singh, M. Daniyal, and P. Dubey, “Detection of phishing attacks in PhiUSIIL dataset using deep learning,” *Procedia Computer Science*, vol. 259, pp. 543–552, 2025, doi: 10.1016/j.procs.2025.04.003.